

HYPERC RESEARCH

SURV-RISK

Toward a Semiparametric Foundation Model for
Risk Inference Under Deep Uncertainty

Directional research paper for HyperC

Position paper and falsifiable research agenda

Organization: HyperC Research, CriticalHop Inc. (d.b.a. HyperC)

Version: 0.1

Date: August 16, 2026

Research status. SURV-RISK is a proposed research program. This paper introduces a formal architecture, testable hypotheses, evaluation protocol, and staged implementation plan. It does not report a trained Large Risk Model or empirical performance results.

Contents

Abstract	2
1. Introduction	2
2. Intellectual Foundations and the Missing Synthesis	4
3. Central Thesis and Formal Problem Statement	6
4. Market Calibration and Survivability-Constrained Valuation	7
5. Universal Risk Representation	9
6. Proposed LRM Architecture	10
7. Synthetic Risk-World Pretraining	14
8. Real-Data Program	15
9. Learning Objectives	16
10. Calibration, Abstention, and Decision Reports	17
11. Evaluation Program	18
12. Falsifiable Hypotheses	19
13. Minimum Viable Research Program	20
14. Scaling Research	21
15. Governance and Alignment Architecture	21
16. Failure Modes and Safeguards	22
17. Open Research Questions	23
18. Research Roadmap	23
19. Expected Scientific Outputs	24
20. Conclusion	24
References	25
Suggested Citation	26

Abstract

Modern quantitative finance is effective when a decision resembles regimes represented in historical data and when the relevant risks are at least partially spanned by observable markets. It is less reliable for projects that combine first-of-kind technology, long time horizons, irreversible capital commitments, sparse analogues, nonlinear dependencies, financing-path constraints, and potentially systemic externalities. In such cases, a conventional scalar risk premium can conceal distinct objects: stochastic outcome risk, parameter uncertainty, structural misspecification, extrapolation uncertainty, market incompleteness, systemic exposure, and normative judgments about unacceptable outcomes.

This paper introduces **SURV-RISK**, a research agenda for a **Large Risk Model (LRM)**: a semi-parametric foundation model with flexible predictive and ambiguity representations. The central hypothesis is that *risk inference itself can be learned as a transferable computational capability*. Rather than fitting a bespoke model for every project, an LRM would be pretrained across heterogeneous real and synthetic “risk worlds” and then perform in-context inference for a new project, its evidence, its dependency graph, and its market environment.

The proposed system returns a structured risk representation rather than a point estimate. Its outputs include a conditional trajectory distribution, an ambiguity set around that distribution, a set of market-compatible stochastic discount factors, an empirical-novelty assessment, systemic and survivability variables, and the expected value of acquiring additional information. The architecture explicitly separates prediction, market valuation, robust decision-making, and governance. Risks priced by existing markets are constrained through no-arbitrage conditions; model uncertainty is represented through context-dependent ambiguity sets; and catastrophic or irreversible outcomes are handled through transparent survivability constraints rather than an arbitrary monetary price for human extinction.

The paper defines the Universal Risk Representation, the LRM architecture, a synthetic risk-world pretraining program, a calibration and abstention policy, and a hierarchy of regime-shift evaluations. The minimum viable scientific test is whether pretraining improves distributional inference across domains while uncertainty expands—rather than contracts—as projects move beyond empirical support. The objective is not universally correct prediction. It is **universally structured uncertainty: a system that communicates what is known, what is priced, what is ambiguous, what is constrained, and what evidence should be acquired next**.

Keywords: foundation models; risk inference; project valuation; distributionally robust optimization; stochastic discount factor; epistemic uncertainty; out-of-distribution detection; systemic risk; survivability; value of information

1. Introduction

Project valuation usually begins with expected cash flows, discount rates, factor exposures, engineering reliability estimates, scenario analysis, Monte Carlo simulation, credit models, and expert judgment. These tools are useful when their statistical and institutional assumptions approximately match the decision environment. The difficult case is not simply a project with a large value on one familiar variable. It is a project whose **joint configuration is unprecedented**.

Consider a capital-intensive project whose expenditure is comparable with historical infrastructure programs but whose combination of construction duration, technological maturity, geopolitical dependencies, supply concentration, demand uncertainty, financing structure,

and irreversible physical consequences has no close precedent. Its capital expenditure may be individually in-distribution while the full project state is outside the support of the available evidence:

$$x_* \notin \text{supp}(\mathcal{D}), \quad \text{even if} \quad \text{CAPEX}_* \in \text{supp}(\text{CAPEX}_{\mathcal{D}}).$$

A conventional model can still return a precise net present value or hurdle rate. Precision, however, is not evidence. When the project lies outside the empirically constrained regime, the central problem is determining how much the data permit anyone to know.

This problem matters for advanced energy systems, biotechnology, artificial-intelligence infrastructure, climate intervention, pandemic resilience, food and water systems, critical minerals, communications infrastructure, and other technologies whose failure modes or externalities may extend well beyond a single investor. It also matters for more ordinary commercial decisions whenever financing, competition, supply chains, and operational dependencies interact in ways absent from historical samples.

SURV-RISK proposes a different object from a larger forecasting model. It seeks to learn a reusable **risk-inference procedure** that can:

1. construct a conditional distribution over project trajectories;
2. retrieve and qualify historical or structural analogues;
3. distinguish aleatory, parameter, model, and deep uncertainty;
4. identify which components of a project can be anchored to market prices;
5. widen ambiguity as empirical support weakens;
6. detect relational or causal breakdowns and abstain when warranted;
7. model systemic spillovers and explicit survivability constraints; and
8. recommend experiments whose information value exceeds their cost.

The key methodological analogy is to foundation models, not to language. A language model is pretrained across diverse sequences and can adapt in context to new tasks. The proposed LRM would be pretrained across heterogeneous **risk worlds**—financial histories, project outcomes, engineering reliability records, macroeconomic regimes, supply networks, simulations, and synthetic stochastic environments. At inference time, the context is a structured body of evidence and the query is a proposed project or action.

The strongest defensible objective is therefore not a universal oracle. It is a **universally applicable risk-calibration architecture whose uncertainty expands when universality fails**.

1.1 Contributions of this paper

This paper makes six directional contributions.

First, it defines risk inference as a transferable learning problem whose output is a structured object rather than a scalar forecast. Second, it proposes a **Universal Risk Representation (URR)** for heterogeneous, time-indexed, provenance-aware project evidence. Third, it specifies a modular LRM architecture that separates nominal prediction, market calibration, ambiguity, survivability, and decision policy. Fourth, it proposes synthetic risk-world pretraining as a way to learn inference behavior despite the scarcity of civilization-scale historical projects. Fifth, it defines an evaluation program centered on calibration under novelty, including delib-

erately missing mechanisms. Sixth, it establishes governance boundaries: prediction, valuation, and acceptable policy must remain distinct and auditable.

1.2 Scope

SURV-RISK is a research program, not a claim that one model can resolve all uncertainty. The system is intended to support high-stakes analysis, not to autonomously authorize investments or certify safety. Its normative thresholds are external governance inputs. Its market layer cannot infer values that markets do not reveal. Its world model cannot identify mechanisms absent from all evidence. A valid output may therefore be an estimate, a wide uncertainty set, or **insufficient evidence**.

2. Intellectual Foundations and the Missing Synthesis

SURV-RISK combines ideas that are individually established but rarely integrated into one auditable architecture.

2.1 Market restrictions without a complete asset-pricing model

Hansen and Jagannathan (1991) showed that security-market data impose nonparametric restrictions on admissible intertemporal marginal rates of substitution and stochastic discount factors. This is important for novel projects because market data can constrain a pricing kernel without requiring one fully specified equilibrium model. Modern machine-learning asset-pricing research further shows that flexible nonlinear models can recover predictive structure and can incorporate no-arbitrage moment conditions into the learning objective. Gu, Kelly, and Xiu (2020) demonstrate the value of nonlinear interactions for empirical risk-premium estimation, while Chen, Pelger, and Zhu (2024) directly use no-arbitrage restrictions in a deep stochastic-discount-factor model.

These results motivate a market-calibration layer, but they do not imply that current markets identify the value of every future state. Markets are incomplete, project payoffs may be unspanned, and future generations or nonmarket ecological states are not fully represented by current asset prices.

2.2 Model misspecification and distributional robustness

Coherent risk measures formalize useful properties of scalar risk functionals, including monotonicity, subadditivity, positive homogeneity, and translation invariance (Artzner et al., 1999). Those axioms do not, by themselves, separate predictive uncertainty from model ambiguity or normative constraints. Robust-control theory instead treats an estimated model as an approximation rather than truth (Hansen & Sargent, 2008). Bayesian calibration of computer models distinguishes uncertain parameters from structural discrepancy between a simulator and reality (Kennedy & O'Hagan, 2001). Distributionally robust optimization (DRO) evaluates decisions over a set of plausible probability measures instead of a single estimated law; Wasserstein formulations provide empirically grounded ambiguity sets and, under suitable assumptions, tractable robust reformulations (Mohajerin Esfahani & Kuhn, 2018; Sinha et al., 2018).

A central limitation in many applications is that the ambiguity set is selected globally or by a tuning procedure disconnected from the local context. Recent work has begun to learn state-dependent Wasserstein radii and nominal scenario distributions, including predictive ambiguity sets optimized for downstream decisions (Guo, 2026). As of August 2026, that result is a new preprint and should be treated as preliminary evidence for adaptive robustness, not as an established solution. It also does not address SURV-RISK's broader problem: cross-domain

transfer, empirical novelty, market calibration, missing mechanisms, systemic spillovers, and explicit survivability constraints.

2.3 Foundation models and learned inference algorithms

Foundation-model research suggests that pretraining across diverse tasks can produce reusable inference behavior, while also creating common-model and governance risks that must be evaluated at the system level (Bommasani et al., 2021). TabPFN is especially relevant because it learns a tabular prediction algorithm across millions of synthetic datasets and performs inference on an unseen dataset in context (Hollmann et al., 2025). Chronos provides a related precedent for probabilistic zero-shot forecasting across heterogeneous time series (Ansari et al., 2024).

SURV-RISK adopts the same high-level strategy but changes the unit of pretraining. A training episode is not one row or one time series. It is a stochastic economic-and-physical world containing projects, markets, dependencies, observation processes, interventions, and reporting biases.

2.4 Calibration, uncertainty under shift, and abstention

Deep ensembles can provide useful empirical uncertainty estimates (Lakshminarayanan et al., 2017), but large-scale evaluations show that the quality of predictive uncertainty often deteriorates under dataset shift (Ovadia et al., 2019). Conformal methods offer finite-sample guarantees for selected marginal risk-control objectives, and Conformal Risk Control extends the framework to monotone loss functions (Angelopoulos et al., 2024). At the same time, exact distribution-free conditional guarantees are generally impossible without assumptions (Barber et al., 2021).

That impossibility is a design constraint rather than an inconvenience. When the required guarantee cannot be supported, the model must not manufacture precision. Abstention is part of the target capability.

2.5 Catastrophic tails, systemic risk, and irreversibility

Rare-disaster asset-pricing research shows that payoffs in catastrophe states can have very different values from payoffs in ordinary states (Barro, 2006). Weitzman (2009) demonstrates how structural uncertainty and weakly identified tails can dominate conventional cost-benefit analysis. Network research shows that tightly connected financial, ecological, technological, and social systems can transmit shocks through nonlinear cascades (Haldane & May, 2011; Helbing, 2013; Gambhir et al., 2025). Real-options theory adds that irreversible investment under uncertainty creates value in waiting, staging, and learning (Dixit & Pindyck, 1994), while value-of-information theory formalizes the benefit of evidence that changes a decision (Howard, 1966).

These strands motivate a model that estimates systemic resilience exposure and value of information, while refusing to collapse human survivability into one hidden scalar utility.

2.6 The gap

The missing synthesis is not another generic forecasting model. It is an architecture combining:

**foundation-model inference + market calibration + learned ambiguity + OOD awareness
+ systemic-risk modeling + survivability constraints + active information acquisition**

The research question is whether these components can be integrated without obscuring their distinct epistemic and normative meanings.

3. Central Thesis and Formal Problem Statement

3.1 Central thesis

The program tests the following hypothesis:

Risk inference can be learned as a transferable computational capability, analogous to in-context learning in foundation models, provided that nominal prediction, market calibration, epistemic uncertainty, structural ambiguity, and normative survivability constraints remain explicitly separated.

A conventional supervised model estimates a task-specific mapping

$$Y = f_{\theta}(X).$$

SURV-RISK instead seeks to learn an operator

$$\mathcal{R} : (a, x, C, D_t, s_t) \mapsto \text{risk representation},$$

where:

- a is a proposed action or staged policy;
- x is the project representation;
- C is the local context and set of analogues;
- D_t is the evidence available at decision time t ; and
- s_t is the market and macroeconomic state.

The target output is

$$\mathcal{R}(a, x) = (\widehat{P}_a, \mathcal{U}_a, \mathcal{M}_t, A_a, S_a, I_a),$$

with the following meanings.

- $\widehat{P}_a$ is a nominal conditional distribution over future trajectories.
- $\mathcal{U}_a$ is a set of plausible alternative distributions representing model ambiguity.
- $\mathcal{M}_t$ is a set of market-compatible stochastic discount factors or pricing kernels.
- A_a is an epistemic assessment containing novelty, analogue quality, disagreement, and support diagnostics.
- S_a is a vector of systemic and survivability effects.
- I_a is a value-of-information object over possible experiments or observations.

A scalar expected return can be derived from this representation only after the decision mandate and admissibility constraints are declared.

3.2 Trajectory-based outcomes

Terminal cash flow is too narrow for the target domain. Let a project trajectory be

$$\tau = (z_0, z_1, \dots, z_T),$$

where z_t may include capital requirements, schedule state, revenues, commodity prices, technical performance, financing conditions, supply failures, regulatory states, environmental variables, and systemic spillovers. The nominal world model estimates

$$\widehat{P}_a(\tau) = P_\theta(\tau \mid a, x, C, D_t, s_t).$$

This representation supports path-dependent default, refinancing risk, staged commitment, option exercise, and irreversible consequences.

3.3 Four uncertainty layers

Every decision report should attempt to separate:

$$\text{total uncertainty} = \text{aleatory} + \text{parameter} + \text{model} + \text{deep uncertainty}.$$

Aleatory uncertainty is randomness conditional on the assumed model. **Parameter uncertainty** concerns unknown quantities inside a selected model class. **Model uncertainty** concerns which structural model is appropriate. **Deep uncertainty** arises when important mechanisms, states, or causal variables may be absent from the candidate family itself.

The categories are not perfectly identifiable in all settings. Their purpose is to prevent a single predictive variance from being misrepresented as a complete account of uncertainty.

3.4 Prediction, valuation, and policy are different objects

The architecture enforces

$$\boxed{\text{prediction} \neq \text{valuation} \neq \text{acceptable policy}}.$$

The predictive layer estimates possible trajectories. The valuation layer weights financial consequences according to a declared pricing or welfare mandate. The governance layer defines outcomes that are inadmissible regardless of modeled return.

4. Market Calibration and Survivability-Constrained Valuation

4.1 Market-compatible pricing set

For traded returns $R_{i,t+1}$, an admissible stochastic discount factor m_{t+1} should approximately satisfy

$$\mathbb{E}_t[m_{t+1} R_{i,t+1}] = 1.$$

Rather than estimate one uniquely correct kernel, SURV-RISK maintains a set

$$\mathcal{M}_t^{\text{market}} = \left\{ m : \left| \widehat{\mathbb{E}}_t[m R_i] - 1 \right| \leq \delta_i, i = 1, \dots, N \right\},$$

where the tolerances δ_i reflect sampling error, liquidity, data quality, and model approximation.

For a project payoff $X_a(\tau)$, the market-consistent valuation set is

$$\mathcal{V}^{\text{market}}(a) = \{ \mathbb{E}_Q[mX_a] : Q \in \mathcal{U}_a, m \in \mathcal{M}_t^{\text{market}} \}.$$

An interval is more honest than a point price when the project is incompletely spanned or when the pricing kernel is weakly identified.

4.2 Context-dependent ambiguity

Let the nominal predictive law be $\widehat{P}_a$. A generic ambiguity set is

$$\mathcal{U}_a = \{ Q : D(Q, \widehat{P}_a) \leq r(a) \},$$

where D may be a Wasserstein distance, an f -divergence, a structured transport cost, or a hybrid discrepancy suited to mixed data.

The radius is project- and context-dependent:

$$r(a) = r_0 + f_\phi(d_{\text{emp}}, d_{\text{struct}}, D_{\text{ensemble}}, E_{\text{retrieval}}, E_{\text{calibration}}, E_{\text{provenance}}).$$

The terms summarize empirical distance, structural novelty, model disagreement, poor analogue retrieval, historical calibration error, and evidence-quality concerns. The architecture should impose monotonic lower-bound constraints so that sufficiently strong movement away from empirical support cannot silently reduce ambiguity.

A simple target property is:

$$A(a_2) \succeq A(a_1) \implies r(a_2) \geq r(a_1) - \eta,$$

for comparable projects where a_2 is demonstrably more novel than a_1 , and η is a small tolerance for measurement noise. A stronger version would require monotonicity along explicitly constructed novelty paths.

4.3 Explicit survivability variables

SURV-RISK does not define human extinction or irreversible civilizational collapse as a dollar loss learned from present markets. Let

$$H(\tau) = (H_1(\tau), \dots, H_K(\tau))$$

be a vector of declared survivability indicators. Candidate dimensions include large-scale mortality, sustained impairment of food production, critical energy-system failure, health-system collapse, irreversible ecological damage, loss of global infrastructure, cascading systemic failures, reduction of recovery capacity, and destruction of future option space.

The admissible action set is

$$\mathcal{A}_{\text{safe}} = \left\{ a : \sup_{Q \in \mathcal{U}_a} Q(H_k(\tau) > b_k) \leq \epsilon_k, \quad k = 1, \dots, K \right\}.$$

The indicator set can draw on earth-system boundary research and systemic-risk frameworks, but it must not mechanically convert those literatures into project-level thresholds (Richardson et al., 2023; Gambhir et al., 2025). The thresholds b_k and tolerances ϵ_k are governance parameters. They must be versioned, attributed, and exposed in every decision report. They are not silently learned from market prices.

4.4 Humanity-Aligned Risk-Adjusted Yield

For an admissible project, define a robust certainty-equivalent value

$$\text{CE}_R(a) = \inf_{Q \in \mathcal{U}_a} \text{CE}_Q(X_a).$$

A provisional **Humanity-Aligned Risk-Adjusted Yield (HARY)** is

$$\text{HARY}(a) = \begin{cases} g(\text{CE}_R(a)), & a \in \mathcal{A}_{\text{safe}}, \\ \text{inadmissible}, & a \notin \mathcal{A}_{\text{safe}}, \end{cases}$$

where g converts robust certainty-equivalent value into an annualized or otherwise comparable statistic when that conversion is meaningful.

The ordering is deliberate:

survivability is a constraint before it is a return adjustment.

This still allows survivability-positive projects to receive credit. A project that produces resources or continuity in catastrophe states can have insurance value even if its ordinary-state return appears modest. The model should therefore estimate **systemic resilience exposure** alongside conventional market beta.

5. Universal Risk Representation

A foundation model requires a common interface across domains. A flat feature vector is insufficient because project risk is relational, temporal, multimodal, and provenance-sensitive.

5.1 Typed temporal property graph

The Universal Risk Representation models a project as a typed graph sequence

$$G_{0:T} = (V_{0:T}, E_{0:T}).$$

Nodes may represent assets, organizations, technologies, geographic regions, commodities, suppliers, counterparties, regulations, financing instruments, ecosystems, populations, and stochastic events. Edges represent ownership, supply, causality, substitution, correlation, financing, physical flow, common exposure, control, contractual obligation, and temporal precedence.

Each node and edge may contain:

- numeric and categorical attributes;
- text and document embeddings;
- time series and event histories;

- geospatial variables;
- probability distributions or expert ranges;
- simulation outputs and fault-tree results;
- action-dependent transition rules; and
- confidence and provenance metadata.

5.2 Evidence provenance as first-class data

An observed value should not be treated identically to a sponsor estimate, a liquid options-implied distribution, or output from an unvalidated simulator. Each evidence item is represented as

$$e_i = (v_i, t_i, s_i, m_i, u_i, c_i),$$

where v_i is the value, t_i the timestamp, s_i the source, m_i the measurement or generation method, u_i the reported uncertainty, and c_i relevant incentives or conflicts.

The model should learn both from evidence and from the historical reliability of evidence classes. Provenance features also enter the ambiguity engine, so conflicting or weakly validated sources can widen the uncertainty set.

5.3 Decision-time snapshots

Every training and evaluation record must preserve the information set that existed at the historical decision date. Later outcomes, revised estimates, and hindsight narratives cannot leak into the model’s context. The corpus therefore requires versioned snapshots:

$$D_t = \{e_i : t_i \leq t\},$$

plus an explicit record of which facts were public, private, delayed, disputed, or unavailable.

5.4 Compositional generalization

The URR should support familiar components in unfamiliar combinations. Evaluation must therefore distinguish marginal familiarity from joint familiarity. Retrieval and novelty diagnostics operate on multiple typed distances rather than one generic embedding:

$$(d_{\text{finance}}, d_{\text{technology}}, d_{\text{scale}}, d_{\text{institution}}, d_{\text{timescale}}, d_{\text{dependency}}, d_{\text{geography}}).$$

A project is not “similar” merely because its capital expenditure or industry label matches a historical case.

6. Proposed LRM Architecture

The architecture contains seven coupled modules with separate outputs, losses, and audit surfaces.

6.1 Multimodal risk encoder

A graph-aware, sequence-aware, set-aware encoder maps project evidence and context into a latent state:

$$h_a = E_\theta(a, x, C, D_t, s_t).$$

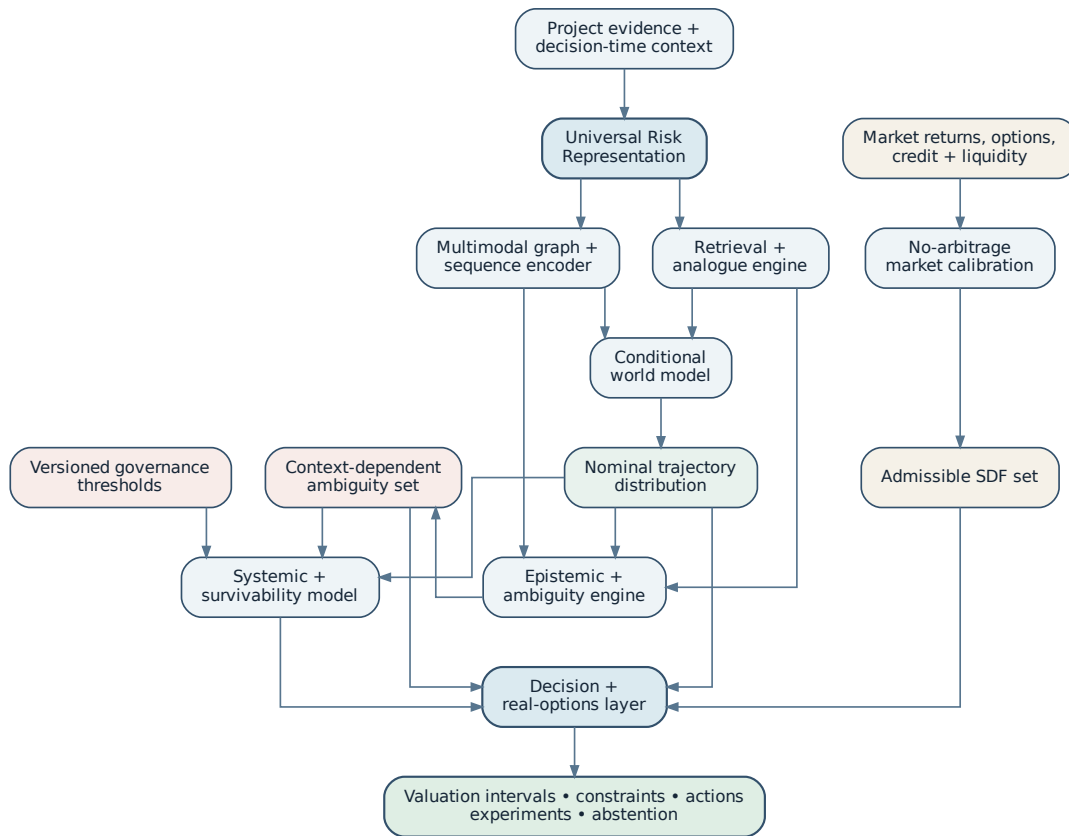


Figure 1. Proposed SURV-RISK Large Risk Model architecture. The modular design keeps nominal prediction, market calibration, ambiguity estimation, survivability constraints, and decision policy separately auditable.

The design must preserve relevant invariances while respecting order where order matters. Candidate components include graph transformers for dependencies, temporal encoders for event histories, set transformers for unordered evidence, and modality-specific encoders for text, tables, geospatial data, and simulation traces.

The encoder should not be optimized only for terminal outcome prediction. Auxiliary tasks should require reconstruction of dependency types, event order, evidence provenance, missingness mechanisms, and causal interventions.

6.2 Retrieval and analogue engine

Before producing a decision estimate, the model retrieves historical, simulated, and structural analogues. Retrieval returns both examples and a diagnostic of where each analogy fails.

For analogue j , the engine produces

$$(w_j, d_j^{(1)}, \dots, d_j^{(K)}, q_j, \Delta_j),$$

where w_j is relevance, the d_j values are typed distances, q_j is evidence quality, and Δ_j describes nontransferable dimensions.

Retrieval quality should be evaluated directly. A persuasive but superficial analogue is a safety failure, not an interpretability success.

6.3 Conditional world model

The world model generates joint trajectories rather than independent forecasts. It may use autoregressive, diffusion, flow-based, latent-state, or simulator-augmented components, provided it returns a coherent distributional representation.

Key requirements are:

- cross-variable dependency and tail-dependence modeling;
- intervention-conditioned trajectories;
- explicit financing and liquidity paths;
- structural-break adaptation;
- compatibility with physical simulators; and
- support for counterfactual staging policies.

The output may be particles,

$$\widehat{P}_a(\tau) \approx \sum_{j=1}^J w_j(a) \delta_{\tau_j},$$

a conditional quantile process, or another flexible generative law.

6.4 Market-calibration layer

The market module learns or constrains stochastic discount factors using traded returns, options, credit spreads, commodity curves, liquidity variables, and macroeconomic states. Its role is to anchor priced risks, not to declare all risks priced.

The module reports:

- pricing errors and their uncertainty;
- the admissible SDF set or posterior;
- spanning quality for the target project;
- decomposition into market-identifiable and unspanned components; and
- sensitivity to test assets and market regime.

A project with weak spanning must receive a wider valuation set rather than an extrapolated point price.

6.5 Epistemic and ambiguity engine

The ambiguity engine estimates how much the nominal law should be distrusted. Inputs include representation distance, retrieval quality, ensemble disagreement, shift diagnostics, simulator disagreement, missingness, evidence provenance, and historical calibration under comparable shifts.

The engine returns both $\mathcal{U}_a$ and a human-readable ambiguity report. Its training objective must balance three competing goals:

1. **coverage or risk control:** plausible realized distributions should not fall outside the set too often;
2. **sharpness:** the set should not be wider than the evidence requires; and
3. **decision relevance:** the set should protect decisions against consequential misspecification, not only average forecast error.

The module should be adversarially tested for **uncertainty inversion**, where novelty increases but the learned ambiguity radius decreases.

6.6 Systemic and survivability layer

This module propagates project trajectories through networked economic, physical, ecological, and social systems. It estimates direct effects, second-order cascades, concentration risk, substitutability, recovery capacity, and state-dependent externalities.

The output includes:

- distributions over declared H_k indicators;
- active survivability constraints;
- systemic-resilience exposure;
- uncertainty attributable to network incompleteness; and
- sensitivity to governance thresholds.

The predictive model may estimate the consequences of a threshold, but it does not select the threshold.

6.7 Decision and real-options layer

The action space is dynamic:

$$\mathcal{A}_t = \{\text{invest, wait, experiment, scale, hedge, redesign, abandon}\}.$$

The robust value of an action is

$$V_R(a) = \inf_{Q \in \mathcal{U}_a} \mathbb{E}_Q[U(a, \tau)],$$

subject to market, financing, and survivability constraints.

For experiment e , the expected value of information is

$$\text{VOI}(e) = \mathbb{E}_{y_e} \left[\max_{a \in \mathcal{A}_{\text{safe}}(D_t, y_e)} V_R(a \mid D_t, y_e) \right] - \max_{a \in \mathcal{A}_{\text{safe}}(D_t)} V_R(a \mid D_t) - \text{Cost}(e).$$

The model should recommend prototype tests, additional measurements, regulatory consultation, supply diversification trials, or other evidence-gathering actions when their robust information value exceeds immediate commitment.

7. Synthetic Risk-World Pretraining

Historical data alone are insufficient because humanity has completed relatively few projects at the scale and novelty of interest. SURV-RISK therefore proposes pretraining over synthetic stochastic environments.

7.1 World generator

For each episode, sample a latent world

$$W \sim \Pi(W).$$

A world specifies:

- a typed dependency graph;
- economic and policy regimes;
- causal mechanisms and nonlinear interactions;
- failure and recovery distributions;
- tail dependence and common shocks;
- financing constraints and default processes;
- technological learning curves;
- physical resource constraints;
- observation and reporting processes;
- market-pricing rules; and
- intervention responses.

The generator produces a decision-time context D_W , candidate actions, partial observations, and subsequent trajectories. Because the latent world is known, the training system can evaluate conditional laws, counterfactual actions, missing mechanisms, and optimal information acquisition.

7.2 Learning an inference algorithm

The model receives incomplete evidence from W and is trained to infer the relevant posterior-predictive structure. In the idealized limit, it approximates

$$P(\tau \mid D_W, a) = \int P(\tau \mid W, a) P(W \mid D_W) dW.$$

The goal is not to memorize one synthetic prior. It is to learn procedures such as analogue selection, regime inference, uncertainty widening, causal qualification, and abstention.

7.3 Hidden-prior risk

A narrow $\Pi(W)$ merely replaces explicit parametric assumptions with hidden generator assumptions. This is one of the program's largest risks. The safeguards are:

- multiple independently designed generator families;
- generator holdouts during evaluation;
- adversarial worlds built to violate training priors;
- transparent generator documentation;
- sensitivity of downstream decisions to generator mix;
- real-data recalibration; and
- external benchmark worlds not controlled by the core development team.

7.4 Missing-mechanism episodes

Some episodes intentionally hide a causal variable. Suppose the true world is

$$Y = f(X, Z),$$

but the model observes only X . During training, Z rarely matters; during a held-out regime, Z dominates. The model cannot infer an unobserved mechanism from no evidence. The target behavior is instead to detect relational breakdown:

Previously stable relationships are failing; the current predictive representation is no longer trustworthy.

A successful system becomes less confident before it becomes completely wrong.

8. Real-Data Program

The real corpus should combine six information classes.

Information class	Representative contents
Financial markets	Returns, yield curves, options, credit, commodities, volatility, liquidity, and cross-asset dependencies
Capital projects	Capital expenditure, schedules, overruns, financing paths, operational performance, failure, delay, and abandonment
Corporate and technology	R&D programs, technology maturity, commercialization, productivity, adoption, competition, and organizational change
Physical and engineering	Reliability records, component failure, maintenance, fault trees, material and energy flows, and simulator outputs
Macrosystem	Trade, food, energy, health, environment, infrastructure, policy, conflict, and network disruptions
Unstructured evidence	Project documents, technical reports, regulatory decisions, scientific literature, risk registers, and expert forecasts

8.1 Selection and reporting bias

Successful projects are often documented more completely than failures, cancellations, and near misses. Sponsor estimates may be strategically optimistic. Engineering incident databases may underreport minor failures. Market data overrepresent tradable risks. These biases should be represented as observation processes rather than treated as random missingness.

8.2 Data rights and reproducibility

The public research program should publish schemas, benchmark splits, synthetic generators, evaluation code, and legally releasable data. Proprietary data can be used for model development only if the public benchmark remains sufficient to evaluate the central claims. Every reported result should include model version, data snapshot, generator version, random seeds, compute budget, and calibration procedure.

9. Learning Objectives

A candidate composite objective is

$$\mathcal{L} = \lambda_p \mathcal{L}_{\text{prediction}} + \lambda_d \mathcal{L}_{\text{distribution}} + \lambda_a \mathcal{L}_{\text{no-arbitrage}} + \lambda_c \mathcal{L}_{\text{calibration}} \\ + \lambda_u \mathcal{L}_{\text{uncertainty}} + \lambda_o \mathcal{L}_{\text{OOD}} + \lambda_g \mathcal{L}_{\text{graph}} + \lambda_s \mathcal{L}_{\text{simulation}} + \lambda_i \mathcal{L}_{\text{information}}.$$

The components serve different purposes.

- $\mathcal{L}_{\text{prediction}}$ scores realized trajectory variables.
- $\mathcal{L}_{\text{distribution}}$ uses proper scoring rules for full distributions and quantiles.
- $\mathcal{L}_{\text{no-arbitrage}}$ penalizes moment violations such as $\hat{\mathbb{E}}[mR] - 1$.
- $\mathcal{L}_{\text{calibration}}$ penalizes undercoverage or excessive risk under comparable regimes.
- $\mathcal{L}_{\text{uncertainty}}$ trains decomposition and ambiguity-set behavior.
- $\mathcal{L}_{\text{OOD}}$ rewards uncertainty expansion or abstention under unsupported extrapolation.
- $\mathcal{L}_{\text{graph}}$ preserves dependency structure and relational consistency.
- $\mathcal{L}_{\text{simulation}}$ transfers physical constraints from simulators.
- $\mathcal{L}_{\text{information}}$ evaluates experiment ranking and posterior decision improvement.

Survivability thresholds are not optimized through this statistical objective. They enter as external constraints during decision evaluation.

9.1 Multi-stage training

A plausible sequence is:

1. representation pretraining on heterogeneous evidence and graph tasks;
2. synthetic-world in-context distributional pretraining;
3. real-data outcome and calibration training;
4. market-module training under no-arbitrage restrictions;
5. ambiguity training with regime and generator holdouts;
6. decision-focused fine-tuning; and
7. conformal or other outer calibration on frozen evaluation protocols.

Joint end-to-end optimization may eventually improve performance, but early versions should preserve modular losses and frozen interfaces to prevent one objective from hiding failure in another.

10. Calibration, Abstention, and Decision Reports

10.1 Calibration under identifiable conditions

Calibration should be evaluated conditionally on meaningful groups and shift classes where sample size permits. Conformal Risk Control can serve as an outer layer for selected monotone losses, while standard probabilistic metrics evaluate nominal distributions. Guarantees must state their assumptions and scope; a marginal guarantee cannot be presented as an individual conditional guarantee.

10.2 Three valid model states

The model has three valid top-level outputs:

estimate, wide uncertainty set, insufficient evidence.

The third is not an operational failure. It is a required behavior when the evidence cannot support a defensible inference.

A provisional abstention rule may combine:

- analogue coverage below a minimum threshold;
- severe model or generator disagreement;
- representation distance beyond validated ranges;
- unresolved evidence contradictions;
- failed conformal or backtest diagnostics;
- missing critical mechanisms indicated by residual structure; and
- extreme sensitivity to plausible ambiguity specifications.

No single indicator is sufficient. Abstention should be calibrated against decision cost, not only classification accuracy.

10.3 Standard decision report

Every major inference should return:

1. the decision-time evidence snapshot and provenance summary;
2. closest analogues and nontransferable differences;
3. nominal trajectory distribution and key dependencies;
4. uncertainty decomposition;
5. ambiguity-set drivers and sensitivity curve;
6. market spanning and pricing-kernel diagnostics;
7. market-consistent valuation interval;
8. active systemic and survivability constraints;
9. recommended action and robust alternatives;
10. highest-value information-acquisition steps; and
11. abstention or escalation conditions.

Explanations should be tied to decision sensitivity. For example:

Demonstrating component reliability for 20,000 operating hours is estimated to reduce the ambiguity radius enough to change the preferred action from “prototype only” to “staged deployment,” provided the supply-concentration constraint is also reduced.

This is more useful than a generic feature-importance chart.

11. Evaluation Program

The principal criterion is **calibration under novelty**, not narrative fluency or average in-distribution accuracy.

11.1 Hierarchy of regime shifts

Evaluation split	Primary question
Random holdout	Can the model interpolate?
Temporal holdout	Does it generalize to future regimes without leakage?
Geographic holdout	Does it transfer across institutions and jurisdictions?
Industry holdout	Does risk-inference behavior transfer across domains?
Scale holdout	Does uncertainty behave sensibly beyond historical project size?
Compositional holdout	Can it handle familiar components in unprecedented combinations?
Structural-break holdout	Does it adapt or abstain when relationships change?
Generator-family holdout	Does it generalize beyond the synthetic priors used in training?
Adversarial world	Does it resist hidden dependencies and misspecification?
Catastrophic-tail simulation	Are extreme-event probabilities and constraints calibrated?
Missing-mechanism test	Does it recognize representation failure?

11.2 Metrics

No single score is sufficient.

Distributional performance should use log score, continuous ranked probability score, Brier score for events, quantile loss, interval coverage, and sharpness.

Market consistency should use pricing errors, no-arbitrage moment violations, stability across test-asset sets, and spanning diagnostics.

Ambiguity performance should measure robust coverage, set size, decision regret, sensitivity to novelty, and the rate of uncertainty inversion.

OOD behavior should measure whether uncertainty, ambiguity radius, or abstention probability rises along controlled novelty paths before accuracy collapses.

Decision performance should use realized regret, probability of ruin, capital survival, utility, financing failure, and comparison with plug-in, Bayesian, fixed-radius DRO, and classical project-finance baselines.

Survivability performance should use robust constraint-violation rates in synthetic worlds where ground truth is known.

Information-acquisition performance should compare the predicted VOI ranking with realized posterior decision improvement after simulated or historical evidence arrival.

11.3 Baselines

The benchmark suite should include:

- linear and generalized-linear project models;
- gradient-boosted trees;
- domain-specific project-finance models;
- Monte Carlo models with manually specified distributions;
- hierarchical Bayesian models;
- Bayesian model averaging;
- deep ensembles;
- fixed-radius Wasserstein DRO;
- context-dependent but non-foundation ambiguity models;
- nearest-analogue methods; and
- task-specific models trained separately in each domain.

The central transfer claim is supported only if a shared pretrained model improves held-out-domain calibration or decision quality without hiding negative transfer.

11.4 Critical unknown-unknown experiment

The missing-mechanism experiment is a gate, not an optional demo. The LRM should be evaluated on how early it recognizes that its model class is failing. Useful measures include detection delay, confidence at failure onset, accumulated decision regret before abstention, and false-abstention rate in ordinary regimes.

11.5 Decision-time backtesting

All historical tests must use only information available at the decision date. The protocol should preregister outcome windows, revised-data handling, project-cancellation treatment, survivorship correction, and transaction or financing assumptions. Apparent performance that disappears under true decision-time reconstruction counts as failure.

12. Falsifiable Hypotheses

The program is organized around six hypotheses.

ID	Hypothesis	Rejection signal
H1	Pretraining across heterogeneous real and synthetic risk environments improves zero-shot or few-shot distributional inference relative to task-specific estimators.	No consistent gain on industry, geography, or generator-family holdouts; systematic negative transfer.
H2	Combining representation distance, analogue quality, ensemble disagreement, and calibration history improves unsupported-regime detection.	A transparent single-indicator baseline performs as well or better across predeclared OOD tests.

ID	Hypothesis	Rejection signal
H3	A context-dependent ambiguity radius constrained by novelty reduces catastrophic overconfidence and robust decision regret.	Learned radii are unstable, shrink under novelty, or fail to outperform fixed-radius and heuristic baselines.
H4	A no-arbitrage-constrained market layer provides useful valuation anchors without requiring a false point price for unspanned projects.	Pricing constraints do not improve valuation stability or produce misleadingly narrow intervals.
H5	Explicit survivability feasibility constraints produce more transparent and stable high-consequence decisions than burying catastrophic outcomes inside a scalar utility function.	Constraint results are dominated by arbitrary threshold sensitivity without informative diagnostics, or scalar alternatives prove equally transparent and robust.
H6	Integrated real-options and VOI analysis improves staged decisions for high-ambiguity projects.	Recommended experiments do not improve posterior decisions after cost, or simple heuristics perform equivalently.

The core thesis is weakened if uncertainty repeatedly decreases as novelty increases, if synthetic pretraining makes decisions primarily reflect hidden generator assumptions, or if cross-domain transfer cannot outperform transparent domain-specific systems.

13. Minimum Viable Research Program

The first prototype should test a narrow and falsifiable question:

Can a pretrained risk model learn when to widen an ambiguity set as a project moves away from historical experience, while preserving useful market-risk calibration?

13.1 Initial domains

The prototype should use three domains with materially different structures:

1. traded financial assets, where market pricing and dense time series are available;
2. conventional capital projects, where cost, delay, financing, and operational outcomes are central; and
3. technology-development programs, where sparse analogues, staged learning, and technical uncertainty dominate.

13.2 Initial outputs

For each case, the model returns

$$\widehat{P}(Y \mid x), \quad r(x), \quad V_{\text{robust}}(x),$$

plus analogue and abstention diagnostics.

The first version need not solve civilization-scale survivability. It should validate the reusable inference backbone, ambiguity behavior, market calibration, and decision-time evaluation pipeline.

13.3 Go/no-go criteria

Before scaling, LRM-0 should demonstrate:

- improved proper-scoring-rule performance on at least two held-out shift types;
- lower robust decision regret than fixed-radius DRO and plug-in optimization;
- monotonic or near-monotonic uncertainty expansion on controlled novelty paths;
- calibrated abstention in deliberately unidentifiable worlds;
- no-arbitrage errors within predeclared tolerances; and
- transparent dependence on synthetic generator families.

Failure on these criteria should trigger redesign or rejection of the broader foundation-model hypothesis before increasing model size.

14. Scaling Research

Model size is an experimental variable, not the objective. Data diversity, world-generator diversity, context length, retrieval capacity, and parameter count should be scaled independently.

The important question is not only whether scale reduces average error. It is whether scale improves:

- analogue selection;
- in-context adaptation;
- uncertainty decomposition;
- recognition of unsupported regimes;
- calibration after structural shift;
- experiment selection; and
- resistance to hidden synthetic priors.

A scaling law for **calibrated epistemic behavior** would be more consequential than a conventional predictive scaling law. It should relate resources to metrics such as OOD coverage, uncertainty inversion rate, abstention quality, and robust decision regret.

15. Governance and Alignment Architecture

15.1 Versioned normative inputs

Survivability indicators, thresholds, probability tolerances, intergenerational assumptions, and social discounting choices must be externally governed and versioned. Every result should identify the active governance profile.

Different institutions may use the same predictive LRM with different normative settings. The system should expose the resulting disagreement rather than average it into hidden weights.

15.2 Separation of institutional roles

A credible deployment separates:

- model developers, who build predictive and uncertainty components;
- market-validation teams, who audit pricing restrictions;

- domain experts, who validate physical and causal representations;
- governance bodies, who select unacceptable-outcome thresholds;
- red teams, who construct adversarial worlds and misuse tests; and
- decision authorities, who remain accountable for action.

15.3 Auditability

Required artifacts include model cards, data statements, generator documentation, calibration reports, threshold registries, decision logs, reproducible evaluation suites, and signed model/version identifiers. High-stakes reports should preserve the complete evidence snapshot and all transformations used to generate the result.

15.4 Limits on automation

The LRM should not autonomously:

- approve civilization-scale deployment;
- define the value of human life or future generations;
- replace legally mandated safety review;
- override explicit constraints because expected return is high; or
- conceal abstention behind an apparently precise executive score.

Its role is to make uncertainty and assumptions more explicit, comparable, and actionable.

16. Failure Modes and Safeguards

Failure mode	Consequence	Primary safeguard
False universality	Precise predictions in unsupported regimes	Typed novelty measures, generator holdouts, abstention
Hidden synthetic priors	Generator design determines conclusions	Independent generators, sensitivity analysis, external adversarial worlds
Market/social conflation	Existing wealth becomes a proxy for humanity	Separate market and survivability layers
Tail underestimation	Catastrophic loss appears negligible	Heavy-tail worlds, stress tests, robust sets, catastrophe-specific metrics
Uncertainty inversion	Confidence rises under extrapolation	Monotonicity constraints and novelty-path tests
Goodhart effects	Projects are optimized to the score	Hidden evaluation suites, causal and physical validation, post-deployment monitoring
Data leakage	Artificial historical performance	Decision-time snapshots and preregistered backtests
Survivorship bias	Failed projects disappear from training	Failure, cancellation, and abandonment corpus
Model feedback	Deployment changes the predicted distribution	Equilibrium and endogenous-policy stress tests
Opaque ambiguity	Robustness becomes an arbitrary penalty	Published set drivers, coverage, and sensitivity curves
Normative capture	One institution defines “humanity alignment”	Plural, explicit, versioned governance profiles

Failure mode	Consequence	Primary safeguard
Cross-domain negative transfer	Shared representation harms specialist tasks	Domain holdouts, routing, adapters, and specialist fallback
Compute centralization	Reproduction and audit become inaccessible	Public smaller models, benchmark generators, and compute reporting

17. Open Research Questions

Several questions remain unresolved and should be treated as first-class research outputs.

17.1 Identifiability of uncertainty decomposition

Aleatory, parameter, structural, and deep uncertainty are conceptually distinct but may not be statistically separable from finite observations. The model should report identification strength and avoid presenting the decomposition as ground truth.

17.2 Geometry of heterogeneous ambiguity sets

A single Wasserstein metric may not respect the semantics of mixed financial, physical, categorical, graph, and temporal variables. The program must study structured transport costs, causal constraints, support changes, and hybrid divergence sets.

17.3 Causal transfer versus correlational transfer

Risk analogues can fail when causal mechanisms differ despite similar observations. The URR and synthetic worlds should support interventions, but causal identification from observational project data will remain limited.

17.4 Equilibrium and reflexivity

If major investors use the same LRM, its recommendations can alter prices, supply chains, project selection, and the distribution it predicts. Deployment evaluation must therefore include endogenous-policy and common-model risk.

17.5 Governance under disagreement

Scientific boundaries, social welfare assumptions, and tolerated catastrophe probabilities will remain contested. The architecture can make disagreement visible; it cannot resolve legitimate political and ethical conflict through prediction.

17.6 Benchmark realism

Synthetic worlds offer known ground truth but may be unrealistic. Historical worlds are realistic but provide incomplete counterfactuals. Progress depends on using both and being explicit about what each can validate.

18. Research Roadmap

Period	Work package	Primary deliverable
Months 1–6	Formal theory and Universal Risk Representation	Formal specification, ontology, benchmark protocol
Months 4–12	Historical and synthetic data program	Versioned multimodal research corpus

Period	Work package	Primary deliverable
Months 7–15	Risk-world generators	Independent synthetic stochastic-environment suites
Months 10–20	Foundation risk backbone	LRM-0 for in-context distributional inference
Months 16–26	Market-pricing module	No-arbitrage-constrained SDF and valuation-set model
Months 20–30	Ambiguity and OOD module	Context-dependent ambiguity and abstention engine
Months 24–34	Systemic and survivability model	Explicit constraint and cascade framework
Months 28–38	Real-options module	Active information-acquisition engine
Months 34–44	Adversarial testing	Public benchmark, external generator holdouts, red-team results
Months 42–48	Integration	LRM-1 research platform, papers, model cards, and reproducible benchmark

The phases overlap intentionally, but no later module should be treated as validated merely because an integrated demonstration is visually convincing.

19. Expected Scientific Outputs

The program is designed to produce separable contributions even if the full LRM hypothesis fails:

1. an open Universal Risk Representation for projects and dependency systems;
2. benchmark datasets with decision-time snapshots and failure cases;
3. synthetic risk-world generators for controlled model-risk experiments;
4. methods for learning context-dependent ambiguity under novelty constraints;
5. a market-consistent foundation-model layer with valuation sets;
6. an OOD benchmark focused on confidence outside historical regimes;
7. a survivability-constrained valuation formalism separating prediction from policy;
8. a value-of-information engine for project design; and
9. governance and audit standards for high-capacity risk models.

This modularity is important. A negative result on universal cross-domain transfer may still leave valuable advances in ambiguity calibration, risk-world generation, or project-data representation.

20. Conclusion

There cannot generally be one universally correct scalar risk adjustment when future states are incompletely observed, markets are incomplete, structural models are uncertain, and some welfare claims are not represented by current market participants.

SURV-RISK therefore seeks to learn three empirical objects together:

world model + ambiguity model + pricing model

while keeping

normative survivability constraints

outside purely empirical calibration.

The intended transformation is from manually specifying a new risk model for every consequential project toward a system that has learned how to construct, criticize, calibrate, and qualify risk models in context. The defining capability is not that the model always knows the answer. It is that the model increasingly knows **how much the available evidence permits anyone to know**, when market information is relevant, when uncertainty must widen, when a decision is inadmissible, and what evidence should be acquired next.

SURV-RISK should therefore be judged by calibration under novelty, robust decision quality, transparent separation of empirical and normative assumptions, and willingness to abstain—not by apparent intelligence in demonstrations.

References

- Angelopoulos, A. N., Bates, S., Fisch, A., Lei, L., & Schuster, T. (2024). Conformal risk control. *International Conference on Learning Representations*. <https://openreview.net/forum?id=33XGfHLtZg>
- Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., et al. (2024). Chronos: Learning the language of time series. *arXiv:2403.07815*. <https://doi.org/10.48550/arXiv.2403.07815>
- Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228. <https://doi.org/10.1111/1467-9965.00068>
- Barber, R. F., Candès, E. J., Ramdas, A., & Tibshirani, R. J. (2021). The limits of distribution-free conditional predictive inference. *Information and Inference*, 10(2), 455–482. <https://doi.org/10.1093/imaiai/iaaa017>
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *The Quarterly Journal of Economics*, 121(3), 823–866. <https://doi.org/10.1162/qjec.121.3.823>
- Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., et al. (2021). On the opportunities and risks of foundation models. *arXiv:2108.07258*. <https://doi.org/10.48550/arXiv.2108.07258>
- Chen, L., Pelger, M., & Zhu, J. (2024). Deep learning in asset pricing. *Management Science*, 70(2), 714–750. <https://doi.org/10.1287/mnsc.2023.4695>
- Dixit, A. K., & Pindyck, R. S. (1994). *Investment under uncertainty*. Princeton University Press.
- Gambhir, A., Albert, M. J., Doe, S. S. P., Donges, J. F., Farajalla, N., Giatti, L. L., et al. (2025). A systemic risk assessment methodological framework for the global polycrisis. *Nature Communications*, 16, 7382. <https://doi.org/10.1038/s41467-025-62029-w>
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5), 2223–2273. <https://doi.org/10.1093/rfs/hhaa009>
- Guo, J. (2026). Learning predictive ambiguity sets for decision-focused distributionally robust optimization. *arXiv:2607.09820*. <https://doi.org/10.48550/arXiv.2607.09820>
- Haldane, A. G., & May, R. M. (2011). Systemic risk in banking ecosystems. *Nature*, 469, 351–355. <https://doi.org/10.1038/nature09659>
- Hansen, L. P., & Jagannathan, R. (1991). Implications of security market data for models of dynamic economies. *Journal of Political Economy*, 99(2), 225–262. <https://doi.org/10.1086/261749>
- Hansen, L. P., & Sargent, T. J. (2008). *Robustness*. Princeton University Press.

- Helbing, D. (2013). Globally networked risks and how to respond. *Nature*, 497, 51–59. <https://doi.org/10.1038/nature12047>
- Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S. B., et al. (2025). Accurate predictions on small data with a tabular foundation model. *Nature*, 637, 319–326. <https://doi.org/10.1038/s41586-024-08328-6>
- Howard, R. A. (1966). Information value theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1), 22–26. <https://doi.org/10.1109/TSSC.1966.300074>
- Kennedy, M. C., & O’Hagan, A. (2001). Bayesian calibration of computer models. *Journal of the Royal Statistical Society: Series B*, 63(3), 425–464. <https://doi.org/10.1111/1467-9868.00294>
- Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in Neural Information Processing Systems*, 30. <https://proceedings.neurips.cc/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html>
- Mohajerin Esfahani, P., & Kuhn, D. (2018). Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming*, 171, 115–166. <https://doi.org/10.1007/s10107-017-1172-1>
- Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J. V., Lakshminarayanan, B., & Snoek, J. (2019). Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems*, 32. <https://proceedings.neurips.cc/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html>
- Richardson, K., Steffen, W., Lucht, W., Bendtsen, J., Cornell, S. E., Donges, J. F., et al. (2023). Earth beyond six of nine planetary boundaries. *Science Advances*, 9(37), eadh2458. <https://doi.org/10.1126/sciadv.adh2458>
- Sinha, A., Namkoong, H., & Duchi, J. (2018). Certifying some distributional robustness with principled adversarial training. *International Conference on Learning Representations*. <https://openreview.net/forum?id=Hk6kPgZA->
- Weitzman, M. L. (2009). On modeling and interpreting the economics of catastrophic climate change. *The Review of Economics and Statistics*, 91(1), 1–19. <https://doi.org/10.1162/rest.91.1.1>

Suggested Citation

HyperC Research. (2026). *SURV-RISK: Toward a semiparametric foundation model for risk inference under deep uncertainty* (Version 0.1). CriticalHop Inc. (d.b.a. HyperC).